Haoming Wang

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Research Profile

PhD candidate (ECE, University of Pittsburgh) working on **multimodal foundation models**, **visual spatial reasoning**, and **interactive embodied AI**. **First-authored CVPR'26 Oral** on a procedural 3D world generator for VLM evaluation; eight additional VLM / diffusion / multimodal papers (published or under review). Direct experience with **LLM-agent driven scene generation, VLM latent representation shaping, sequential decision-making over visual observations (key-frame and landmark selection), and Tree-of-Thought planning for embodied navigation** – all immediately relevant to RL training systems for multimodal foundation models and instruction-conditioned world models.

Education

Ph.D., Electrical and Computer Engineering, University of Pittsburgh — **Sept 2022 – May 2027 (Anticipated)**

Advisor: Prof. Wei Gao, Intelligent System Lab. Focus: multimodal LLMs / VLMs, spatial reasoning, embodied AI.

B.Eng., Automation, Zhejiang University — GPA 3.8/4.0 — Sept 2018 – May 2022
Chu Kochen Honors College.

Selected Publications – Multimodal Foundation Models & Visual Reasoning

[CVPR'26 Oral] InfiniBench: Infinite Benchmarking for Visual Spatial Reasoning with Customizable Scene Complexity — CVPR 2026

Haoming Wang, Qiyao Xue, Wei Gao. *Acceptance 25.4%; selected for Oral.*
[arXiv:2511.18200] [Code] [Data]

- LLM-agent (Gemini-2.5-Pro) translates natural-language scene specs into procedural constraints and refines them via a CoT feedback loop driven by BEV collision maps – a recipe for wrapping LLMs around constrained generators with verifiable reward, directly relevant to instruction-conditioned world generation.
- Cluster-based layout optimizer + frontier-based task-aware camera trajectory planner render **photo-realistic egocentric videos** (Blender Cycles) of theoretically unlimited 3D scenes with parameterized compositional / relational / observational complexity.
- Diagnostic study across Gemini-2.5-Pro, GPT-5, LLaVA-Video-7B, InternVL3.5-8B, Cambrian-S-7B reveals BEV vs. egocentric perspective gaps and complexity-driven failure modes.

Uncovering and Shaping the Latent Representation of 3D Scene Topology in Vision-Language Models — 2026 (Preprint)

Haoming Wang, Wei Gao. [Paper] [Code]

Identify a VLM latent subspace aligned with Laplacian eigenmaps of the 3D scene graph; a **Dirichlet-energy regularizer** beats SFT by up to **12.1%** – a representation-level objective for hierarchical-latent / RL post-training.

MosaicThinker: On-Device Visual Spatial Reasoning for Embodied AI via Iterative Construction of Space Representation — 2026 (Under Review)

Haoming Wang, Qiyao Xue, Weichen Liu, Wei Gao. [arXiv:2602.07082]

Training-free cross-frame spatial reasoning for small on-device VLMs; **Gaussian-kernel iterative key-frame selection** as a sequential information-acquisition policy under an observation budget. Up to **40%** accuracy gain on VSI-Bench / Metro-Spatial-QA; deployed on Jetson Orin, AR Glass, smartphone.

ReMindView: Reasoning Path and Latent State Analysis for Multi-view Visual Spatial Reasoning — 2026 (Under Review)

Qiyao Xue, ***Haoming Wang*** (co-author), Weichen Liu, Shiqi Wang, Yuyang Wu, Wei Gao. [arXiv:2512.02340]

Cognitively grounded multi-view benchmark (>50K VQAs across 15 VLMs); linear-probe + entropy-dynamics analysis yields an uncertainty signal that separates correct vs. incorrect reasoning trajectories – a candidate reward for RL post-training.

Spatial Reasoning in Multimodal Large Language Models: A Survey of Tasks, Benchmarks and Methods — 2026 (Under Review)

Weichen Liu, Qiyao Xue, ***Haoming Wang***, Xiangyu Yin, Boyuan Yang, Wei Gao.

Embodied AI & Interactive Multimodal Systems

HEVEN: Small Object Search and Navigation on VLM-Based Autonomous Mobile Systems 2026 (Under Review)

Anonymous Authors (incl. ***Haoming Wang***, 3rd author). *Double-blind under review.*

Spatial Semantic Tree + **Tree-of-Thought** planner with asynchronous speculative execution that pipelines cloud VLM reasoning with a >20 Hz reactive controller; **73.86%** SR (3× over DOVSG), 8.7× memory and 4.4× inference reduction.

GlobalNav: Daily Object Navigation in VLM-Based Mobile Systems with Aligned Local and Global Views 2026 (Under Review)

Anonymous Authors (incl. ***Haoming Wang***, 3rd author). *Double-blind under review.*

Infrastructure-assisted VLM navigation aligning CCTV global views with egocentric robot views via 8-bin spatial topology + spectral-graph matching; GPT-5.2 CoT hierarchical planning with Voronoi-waypoint fallback. **90.91%** SR (vs. 53.82% best baseline) on 1500 Habitat episodes; ∼85% on real LIMO + Jetson Orin testbed.

Diffusion & Generative-Model Interpretability

Deciphering Personalization: Towards Fine-Grained Explainability in Natural Language for Personalized Image Generation Models (FineXL) 2025 (Under Review)

Haoming Wang, Wei Gao. [Paper]

Training-free decomposition of CLIP-space divergence between base and personalized **diffusion / GAN / autoregressive** generators into orthogonal VLM-discovered concepts; cuts single-aspect MAE **56%**, multi-aspect selection error ≥**50%**. Validated on SD v2.1, SD 3.5, ControlGAN, Anole-7B.

FreezeAsGuard: Mitigating Illegal Adaptation of Diffusion Models via Selective Tensor Freezing 2024 (Preprint)

Kai Huang, ***Haoming Wang*** (co-author), Wei Gao. [Paper] [Code]

Bilevel optimization learns a tensor-freezing mask over Stable Diffusion via sigmoid relaxation – a differentiable-mask recipe for diffusion training; 37–47% stronger mitigation than UCE / IMMA, 48% GPU memory and 21% wall-clock savings.

Other First-Author Publications

[MobiSys'25] Never Start from Scratch: Expediting On-Device LLM Personalization via Explainable Model Selection MobiSys 2025

Haoming Wang, Boyuan Yang, Xiangyu Yin, Wei Gao. *(18.0% acceptance.)* [Paper]

[MobiCom'25] When Device Delays Meet Data Heterogeneity in Federated AIoT Applications MobiCom 2025

Haoming Wang, Wei Gao. *(17.1% acceptance.)* [Paper]

[AAAI'25] Tackling Intertwined Data and Device Heterogeneities in Federated Learning with Unlimited Staleness AAAI 2025

Haoming Wang, Wei Gao. *(23.4% acceptance.)* [arXiv]

Research Experience

Graduate Student Researcher, Intelligent System Lab, University of Pittsburgh Sept 2022 – Present

- **(May 2025 – Present)** Multimodal foundation models for visual spatial reasoning: LLM-agent + procedural 3D world generator (CVPR'26 Oral); on-device cross-frame spatial reasoning via sequential key-frame policies; representation-level latent shaping of VLMs.
- **(Nov 2024 – May 2025)** Generative-model interpretability and on-device LLM personalization (FineXL, Xpert); training-free fine-grained explanation of diffusion / GAN / autoregressive image generators.
- **(Sept 2022 – Oct 2024)** Federated learning systems with intertwined data and device heterogeneity (MobiCom'25, AAAI'25); large-scale distributed training infrastructure across heterogeneous clients.

Awards & Recognition

ECE Research Assistant of the Year, University of Pittsburgh April 2026

Department of Electrical and Computer Engineering; recognized at the EGSO event.

Technical Skills

Models: Stable Diffusion, CLIP, Gemini, GPT-4o/5, InternVL, Qwen-VL, LLaVA-Video, SpatialVLM, Grounding-SAM, DepthPro.

Simulation & Edge: Habitat, Isaac Sim, Infinigen, Blender; Jetson Orin, AR Glass, Android (ONNX / NNAPI).

Programming: Python, PyTorch, CUDA, C++.